

ME 311 — Fluid Mechanics

Lecture · Fall 2026 — Term 261

Course code	ME 311
Component	Lecture
Credit hours	3 (3-0-1)
Prerequisite	ME 231 Thermodynamics I
Textbook	Cengel & Cimbala, Fluid Mechanics: Fundamentals and Applications, McGraw-Hill.

Course description

Properties of fluids; fluid statics and pressure measurement; the control-volume forms of mass, momentum and energy; the Bernoulli equation; dimensional analysis and similitude; viscous flow in pipes and head losses; external flow, drag and lift; and an introduction to turbomachinery.

Learning outcomes

- Compute hydrostatic forces on submerged plane and curved surfaces.
- Apply conservation of mass, momentum and energy to control volumes.
- Use dimensional analysis and similitude to plan and interpret model tests.
- Size pipes and pumps by evaluating major and minor losses in internal flow.
- Estimate drag and lift on bodies in external flow.

Weekly plan

Week	Topic
1-2	Introduction, fluid properties, viscosity
3-4	Fluid statics, manometry, forces on surfaces, buoyancy
5	Fluid kinematics and the Reynolds transport theorem
6-7	Mass, Bernoulli and energy equations
8	Midterm examination
9	Momentum analysis of flow systems
10	Dimensional analysis and modeling
11-12	Internal flow: laminar and turbulent pipe flow, losses, piping networks
13	External flow: boundary layers, drag and lift
14	Introduction to turbomachinery



15 Review and final examination

Assessment

Component	Weight
Homework and quizzes	20%
Laboratory reports	10%
Midterm examination	30%
Final examination	40%